



Validity



Roadmap

Validity

Ways to Organize Validity

Causal Inference Perspective

- Cooke & Campbell
 - External Validity
 - Internal Validity
- Hill's Criteria

Education Measurement Perspective

- Content-related Validity
- Criterion-related Validity
- Construct-related Validity

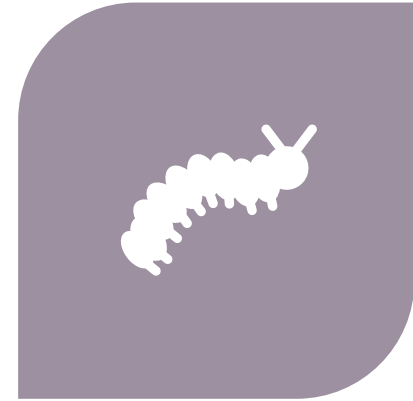
Validity Definitions



The extent to which a test measures what it was designed to measure.



Agreement between a test score or measure and the quality it is believed to measure.



Proliferation of definitions led to a dilution of the meaning of the word into all kinds of “validities”

Validity That Isn't

Face Validity

- The mere appearance that a test has validity.
- Does the test look like it measures what it is supposed to measure?
- Do the items seem to be reasonably related to the perceived purpose of the test.

Does a depression inventory ask questions about being sad?

- Not a “real” measure of validity, but one that is commonly seen in the literature.
- Not considered legitimate form of validity by the Joint Committee.

Ways to Organize Validity

- Causal Inference Perspective
 - Shadish, Cook, and Campbell
 - Different types of validity
- Education Measurement Perspective
 - *Standards for Educational and Psychological Testing*
 - Different types of evidence



This Time...
Causal
Inference
Perspective

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Causal Inference Perspective

- Internal validity
 - Some think of it as:
 - Cause and effect in experimentation;
 - high levels of control;
 - elimination of confounding variables
 - Can also be thought of as the claim/finding from a specific study
- External validity
 - to what extent one may safely generalize the (internally valid) finding
 - (a) from the sample studied to the defined target population and
 - (b) to other populations (i.e. across time and space).
Generalize to other people



Internal Validity

- Differs by kind of study
 - For experiments....
 - For quasi-experiments...
 - For observational studies...
- General idea is that your finding is “real”

External Validity

To what extent one may safely generalize?

- the (internally valid) finding from the sample studied to the
 - defined target population and
 - to other populations
 - (i.e. across time and space).

Generalize to other people



Types of External Validity

Population validity

- can the sample results be generalized to the target population

Ecological validity

- whether the results can be applied to real life situations.
- Generalize to other (real) situations

Generalizability

The burden of proof

- Avoid simplistic generalizations to members of other cultures and people in different times (including differences)
- Those who question the generalizability of a study
 - should propose **when, how, and why** it is not generalizable

Personally, my default assumption is that the study doesn't generalize

- And that it is up to the authors of the study to convince the reader (*me*)
- But, that is not the norm...

Thinking About Generalizability

Slightly less than arbitrary rules of thumb:

- My quick rule of thumb:
 - if you *could* have participated in the study,
 - then the results mostly likely generalize to you
- My thoughtful rule of thumb:
 - If the reasons you *couldn't* have participated is unrelated to findings of the study,
 - then the results probably generalize to you.

Assumption: The study is internally valid.

This approach can be applied to groups other than yourself...

Example: British Doctors' Study

- Does smoking cause cancer?
 - Wasn't obvious or accepted in the medical community before this study
 - Doctors were often associated with tobacco ads
 - (like this one from 1946)
- Doll and Hill (1954, 1956) tracked 40,701 Male British Doctors
 - (2/3 of all registered UK physicians in Oct 1951)

It does.



According to a recent Nationwide survey:
**MORE DOCTORS SMOKE CAMELS
THAN ANY OTHER CIGARETTE**

DOCTORS in every branch of medicine—113,597 in all—were queried in this nationwide study of cigarette preference. Three leading research organizations made the survey. The gist of the query: What cigarette do you smoke, Doctor?
Brand named most was Camel!
The rich, full flavor and cool mildness of Camel's blend of costlier tobaccos seem to have the appeal to the smoking tastes of doctors as to other smokers. If you are a Camel smoker, this preference among doctors will hardly change. If you're not—well, try Camels now.



Your "T-Zone" Will Tell You...

**T for Taste . . .
T for Throat . . .**
that's your
proving ground
for any cigarette.
See if Camels
don't suit your
"T-Zone" to a "T."

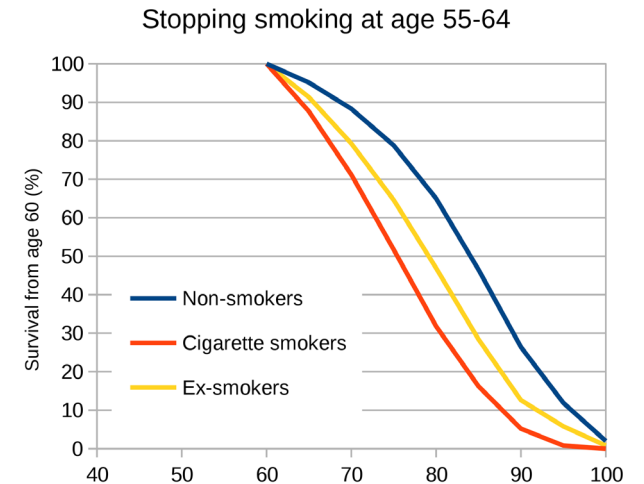
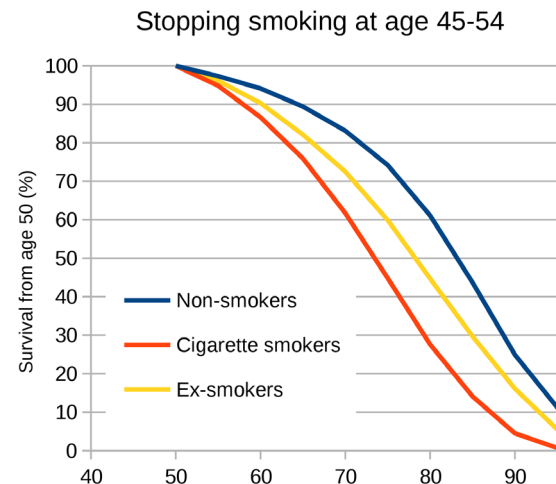
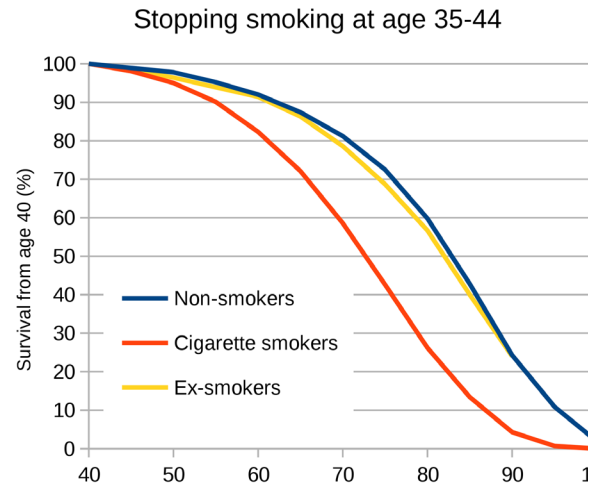
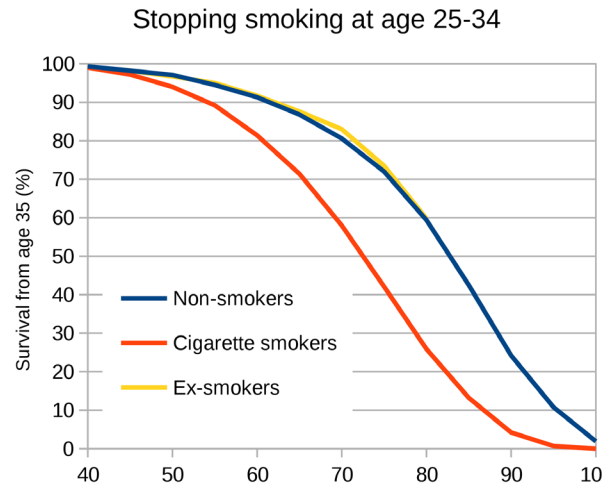


CAMELS *Costlier Tobaccos*

British Doctors' Study (1951-2004)

- Doll and Hill (1954, 1956) tracked 40,701 Male British Doctors
 - (2/3 of all registered UK physicians in Oct 1951)
 - They examined whether there was a relationship between smoking and cancer/respiratory disease.
 - They compared non-smokers, ex-smokers, and current smokers by age.
- Their studies changed the conversation about cancer and smoking

British Doctors' Study (1951-2004)



British Doctors' Study and those slightly less than arbitrary rules of thumb

- *Could I* have participated in the study?
 - No
 - Why Not?:
 - Not British,
 - Not a Man,
 - Not an MD,
 - Born after this study was published
- Are those reasons related to the study's finding?
 - Probably not.
- Verdict: Don't Smoke



If you want to learn more about smoking, statistics, causation, or related history...



<https://www.cancer.org/latest-news/the-study-that-helped-spur-the-us-stop-smoking-movement.html>



<https://www.history.com/news/cigarette-ads-doctors-smoking-endorsement>



<http://tobacco.stanford.edu>

(Tobacco Advertising)



<https://www.industrydocuments.ucsf.edu/tobacco/>

(Tobacco Industry Archives)

Wrapping up...
Causal
Inference
Perspective

This Time... Education Measurement Perspective





Key Ideas

- The validity is based on accumulated evidence
- Validity of past inferences changes based on the current available evidence
- Evidentiary bases for interpretations can be derived through a variety of methods

Education Measurement Perspective

There is a total mess of “validities” and their definitions, what to do?



1985 - Joint Committee of

AERA: American Education Research
Association

APA: American Psychological
Association

NCME: National Council on
Measurement in Education



developed *Standards for Educational and Psychological Testing (revised in 1999)*.

Standards for Educational and Psychological Testing



According to the Joint Committee:



Validity is the evidence for inferences made about a test score.



Three types of evidence:

Content-related
Criterion-related
Construct-related



Different from the notion of “different types of validity”

Three types of evidence

Content-related evidence

- Content Validity

Criterion-related evidence

- Criterion Validity

Construct-related evidence

- Construct Validity



Wrapping up...
Education
Measurement
Perspective



This Time...
Content
Validity

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Big Idea

- Content validity is
 - "the degree to which elements of an assessment instrument are relevant to and representative of the targeted construct for a particular assessment purpose." (Haynes, 1995)



Content validity

- When trying to measure a domain are all sub-domains represented?
 - Complimentary to
 - domain sampling theory and
 - reliability
- However, increasing content validity lowers internal consistency reliability

Main steps to establishing evidence



Define the content
domain...



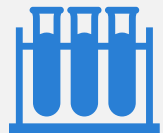
Outline...



Subject matter
experts evaluate...



Does the test adequately sample the content or behavior domain that it is designed to measure?



If items are not a good sample, results of testing will be misleading.



Usually developed during test development.

Not generally empirically evaluated.
Judgment of subject matter experts.

Content-Related Evidence

Content-Related Evidence



To develop a test with high content-related evidence of validity, you need:

good logic
intuitive skills
Perseverance



Must consider:

wording
reading level

Threats to content validity

Other content-related evidence terms

Construct underrepresentation:

- failure to capture important components of a construct.
 - Test is designed for chapters 1-10 but
 - only chapters 1-8 show up on the test.

Construct-irrelevant variance:

- occurs when scores are influenced by factors irrelevant to the construct.
 - Test is well-intentioned, but problems secondary to the test
 - negatively influence the results
 - (e.g., reading level, vocabulary, unmeasured secondary domains)



Wrapping up...
Content
Validity

This Time... Criterion- Related Validity



Criterion-Related Validity

the success of measures used for prediction or estimation.



Two types

Concurrent
validity

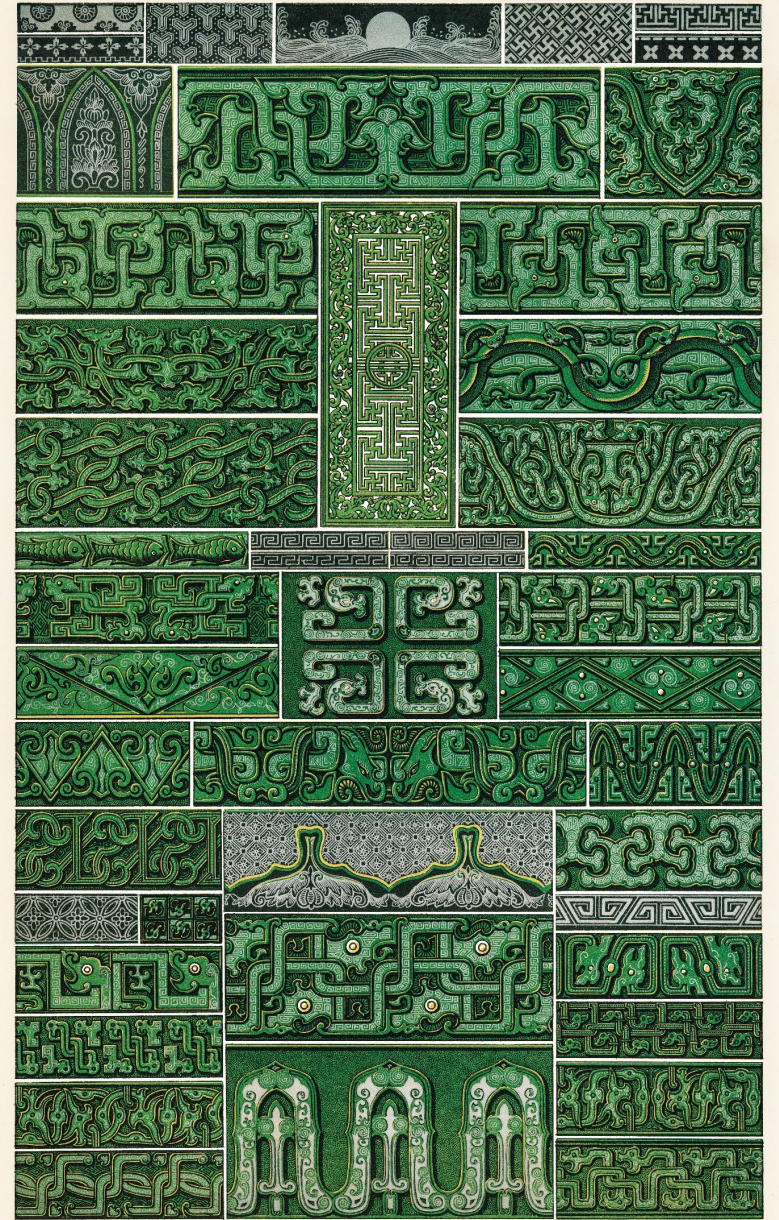
Predictive
validity

Concurrent validity

- the degree to which a test correlates with
 - external criteria that is
 - measured at the same time
- (e.g., does a depression inventory correlated with clinical diagnoses)

Predictive validity

- the degree to which a test predicts (correlates with)
 - an external criteria that is
 - measured some time in the future



Criterion-Related Evidence

Tells us how well a test corresponds with a particular criterion

criterion: behavioral or
measurable outcome

SAT predicting GPA (GPA is
criterion)

BDI scores predicting suicidality
(suicide is criterion).



Used to “predict the future” or “predict the present.”

Criterion-Related Evidence

Predictive Validity Evidence

forecasting the future

how well does a test predict future outcomes

SAT predicting 1st yr GPA

most tests don't have great predictive validity



decrease due to time & method variance

Criterion-Related Evidence

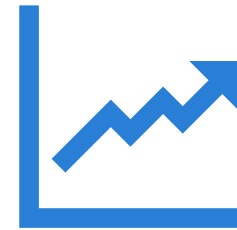


Concurrent Validity Evidence

forecasting the present

how well does a test predict current similar
outcomes

job samples, alternative tests used to
demonstrate concurrent validity evidence



generally higher than predictive
validity estimates

Quantifying Criterion- Related Evidence

Validity Coefficient

- correlation between the test and the criterion

However,

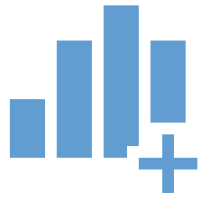
- recall that r^2 indicates explained variance.

Sound Problematic??

Evaluating Criterion- Related Evidence

Look	Look for changes in the cause of relationships
Examine	Examine what the criterion really means. • Optimally the criterion should be something the test is trying to measure
Review	Review the subject population in the validity study.

Evaluating Criterion-Related Evidence



Ensure the sample size in the validity study was adequate.



Never confuse the criterion with the predictor.



Watch for restricted ranges!

Evaluating Criterion-Related Evidence



Review evidence for validity
generalization.

Tests only given in laboratory settings, then
expected to demonstrate validity in classrooms?

Ecological validity?



Consider differential prediction.

External validity?



Wrapping Up... Criterion-Related Validity



This Time... Construct Validity

Construct Validity

Are you measuring what you are intending to measure, overall?

Specific types...

- Intentional validity
- Representation validity or translation validity
- “Face validity”
- Observation validity
- Convergent validity
- Discriminant validity

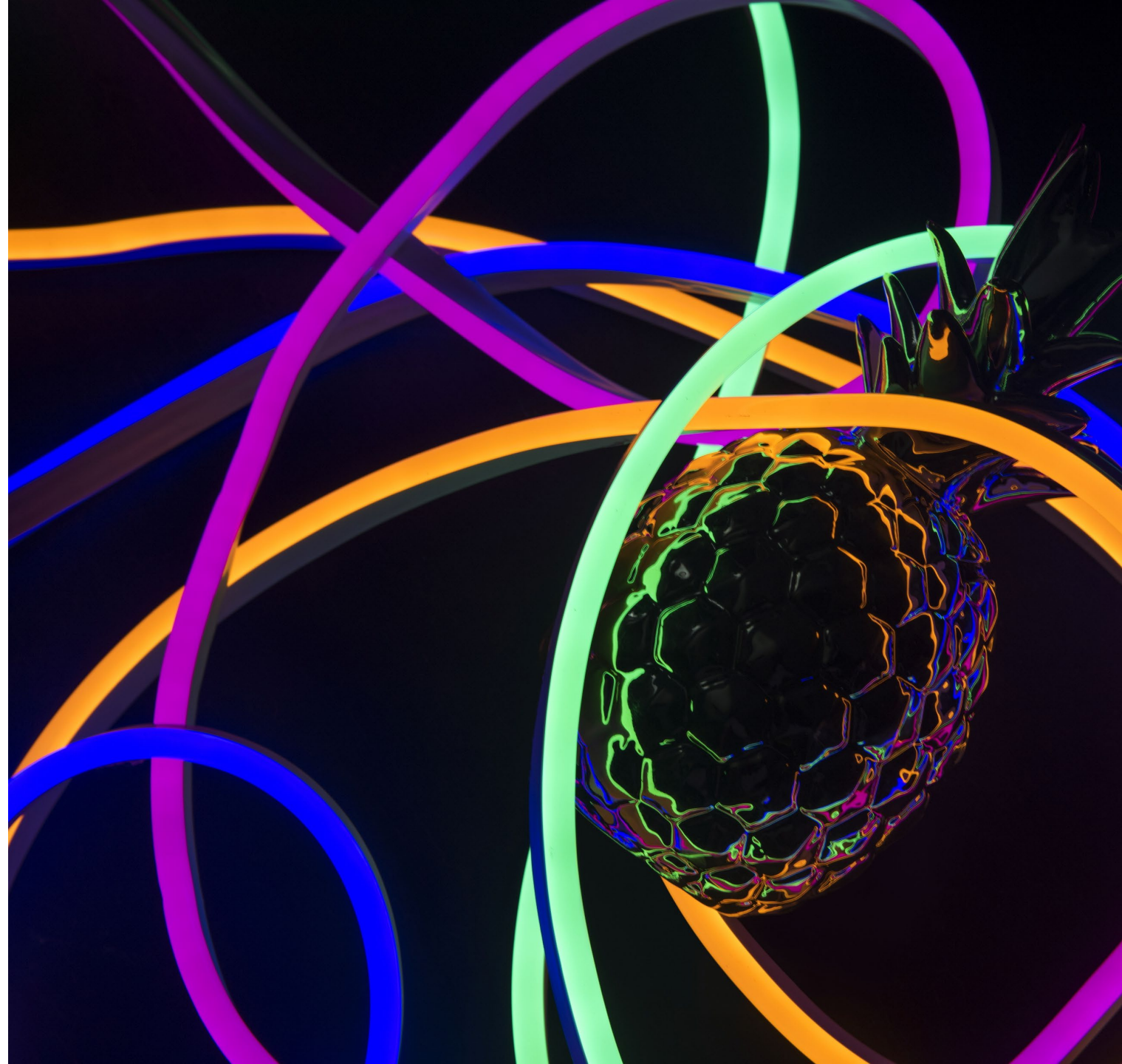
Construct Validity

Intentional validity

- are you measuring what you are intending
 - and not something else?
- Requires ...

Representation validity

- aka ...
- How well have the constructs been translated into measurable outcomes?





Construct Validity

Observation validity

- How good are the measures themselves?

Convergent validity

- refers to the degree to which a measure is correlated ...

Discriminant validity

- describes the degree to which the operationalization does...

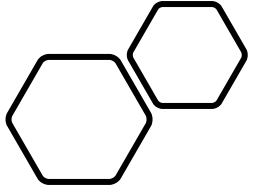
Construct-Related Evidence

Construct: something constructed by mental synthesis

- What is Intelligence? Love? Depression?

Construct Validity Evidence

- Assembling evidence about what a test means
 - (and what it doesn't)
- Is a sequential process



Construct-Related Evidence



Convergent Evidence

obtained when a measure correlates ...

- Self-report, collateral-report measures...



Discriminant Evidence

obtained when a measure...



Wrapping up...
Construct Validity

This Time... Connecting Validity to Reliability



Recall

- The **standard error of measurement** (SEm) estimates how repeated measures of a person on the same instrument tend to be distributed around their “true” score.
 - The true score is always an unknown because ...

Validity & Estimates

- Standard Error of Estimate:
 - $s_{est.} = s_{Y-\hat{Y}} = s_y \sqrt{(1 - r^2) \left(\frac{N-1}{N-2} \right)}$
 - $s_{est.}$ standard error of estimate
 - s_y standard deviation of the test
 - r validity of the test
- Essentially, regression all over again.



Maximum Validity

- Maximum Validity depends on Reliability
 - $r_{12\max} = \sqrt{r_1 r_2}$
 - $r_{12\max}$ is the maximum validity
 - r_1 is the reliability of test 1
 - r_2 is the reliability of test 2
- 

Maximum Validity

Reliability of Test	Reliability of Criterion	Maximum Validity (Correlation)
1	1	1.00
0.8	1	0.89
0.6	1	0.77
0.4	1	0.63
0.2	1	0.45
0	1	0.00
1	0.5	0.71
0.8	0.5	0.63
0.6	0.5	0.55
0.4	0.5	0.45
0.2	0.5	0.32
0	0.5	0.00
1	0.2	0.45
0.8	0.2	0.40
0.6	0.2	0.35
0.4	0.2	0.28
0.2	0.2	0.20
0	0.2	0.00
1	0	0.00
0.8	0	0.00
0.6	0	0.00
0.4	0	0.00
0.2	0	0.00
0	0	0.00



Wrapping up... Connecting Validity to Reliability